

Sensitivity-aware inspection prioritisation for archaeological landscapes: a reproducible Taxila observatory

Abstract

Conservation teams need to prioritise inspections across archaeological landscapes using environmental observations collected at different spatial and temporal scales. We present the Climate-contextual Heritage Inspection Prioritisation (CHIP) framework and evaluate how its recommendations change under alternative indicator, spatial-support and uncertainty assumptions at the Taxila World Heritage property, Pakistan. The framework combines 55 Landsat scenes across five epochs, approximately 30 m terrain data, daily weather for 1991–2025 and 17 geolocated heritage components. A hierarchical landscape–terrain score is evaluated through domain and factor ablation, three neighbourhood radii, 27 structural scenarios, spatial-block resampling, coordinate perturbation and joint spatial–decision simulation. Of 386,280 common-support cells, 16.93% met at least two adverse spectral criteria. Removing MNDWI changed rank concordance to $\rho = -0.280$, whereas removing NDVI or NDBI retained $\rho > 0.989$, showing that information content was unevenly distributed across indicators. Giri led the fixed 500 m ranking and entered the top three in 73.24% of joint score vectors; its 95% rank interval nevertheless spanned 1–9. Spatial reconstruction and decision assumptions produced distinct inspection rationales, particularly for Bhallar. Opposing precipitation trends in Open-Meteo and NASA POWER limited climatic attribution. In a separate seven-family spatial land-cover comparison, development-selected random forest achieved macro-F1 0.821 (95% interval 0.691–0.835); feature controls and calibration experiments exposed limitations that a single accuracy score would conceal. CHIP translates heterogeneous observations into component-specific inspection priorities while quantifying the conditions under which those priorities change. Its contribution is an experimentally evaluated decision-support procedure for landscape screening, with monument-condition validation remaining a subsequent field task.

Keywords: cultural heritage; Taxila; remote sensing; spatial uncertainty; inspection prioritisation; reproducibility.

1 Introduction

Archaeological landscapes are monitored through observations that differ in spatial scale, temporal coverage and relevance to conservation. A satellite pixel describes a land surface; a weather grid describes atmospheric conditions over a much larger area; a terrain model approximates pathways that may concentrate runoff. None directly observes the condition of a wall, stupa or buried deposit. Yet conservation teams must use incomplete evidence to decide where to inspect first. The practical challenge is to translate landscape observations into a defensible inspection sequence while retaining the limits of each observation.

Remote sensing and geographical information systems have documented development pressure around heritage clusters and supported multi-hazard assessments [1–3]. At Taxila, Khan et al. combined ground-based and satellite evidence to assess risks to World Heritage components [4]. That work establishes the local conservation problem and the importance of field observations. A further question concerns the stability of an inspection decision: would the same components remain priorities if the analyst changed the neighbourhood radius, removed a

correlated indicator, or admitted uncertainty in spatial sampling? An assessment that supplies only a final map cannot answer this question.

Climate adds a second interpretive difficulty. Heritage exposure depends on material, maintenance, drainage and microenvironment, as well as meteorological forcing [5–7]. Annual weather summaries may describe conditions different from those prevailing before the satellite scenes used to construct an epoch. Gridded products may also disagree. Climate context therefore requires explicit alignment to acquisition dates and comparison across products before it can inform a conservation explanation. Five multi-year satellite composites do not provide sufficient temporal replication to identify a causal weather effect on archaeological condition.

Model evaluation presents a related problem. Nearby pixels can share both predictors and labels, making random partitions an optimistic representation of geographic transfer [8–10]. Spatial separation addresses part of this dependence, but it cannot turn a global land-cover product into independent field truth. A classifier that reproduces WorldCover labels may recognise vegetation or built surfaces without establishing which heritage component is deteriorating. This distinction remains necessary as model families become more capable.

We address these issues through an evidence-to-decision workflow, CHIP, that retains separate landscape, terrain, climate and proxy-evaluation paths. The methodological contribution is to evaluate the decision rule itself: indicator ablation identifies which observations control the ranking; spatial and positional experiments test dependence on sampling support; and crossed spatial–decision simulations quantify their combined effect. A separate seven-family land-cover comparison evaluates the spectral representation under geographic separation. These experiments connect component-level priorities to the evidence and assumptions that produce them, allowing inspection teams to distinguish broadly persistent priorities from priorities associated with a particular environmental mechanism.

2 Research aim

The study aims to identify which Taxila components merit early field inspection and which parts of that ordering depend on analytical choices. Specifically, it examines whether adverse spectral changes converge on common spatial support; whether weather interpretation changes with acquisition window or data product; whether component ranks remain stable across neighbourhood radii, indicator composition and resampling; and whether land-cover proxy agreement persists under buffered spatial evaluation. These questions link the computational evidence to a practical conservation task while keeping landscape screening distinct from direct assessment of heritage condition.

3 Materials and methods

3.1 Study area, inventory and evidence sources

Taxila is a serial World Heritage property in Punjab, Pakistan, inscribed under reference 139 [11]. The inventory contained 18 component records. Seventeen had usable point coordinates; Saraikala (139-002) remained in the documentary inventory but was excluded from coordinate-dependent analysis because its location was unresolved. All metric operations used WGS 84/UTM zone 43N (EPSG:32643). Point-centred circles of 250, 500 and 1,000 m defined analytical neighbourhoods, with 500 m used for primary reporting. They are neither official property polygons nor statutory or UNESCO buffer zones (Fig. 1).

Landsat Collection 2 Level-2 surface reflectance supplied five post-monsoon three-year composites: 2003–2005, 2008–2010, 2013–2015, 2018–2020 and 2023–2025, abbreviated E2004 to E2024. They contained 11, 10, 8, 8 and 18 retained scenes, respectively. Landsat 7 supplied the first two epochs, Landsat 8 the middle two, and Landsat 8/9 the final epoch. Source records

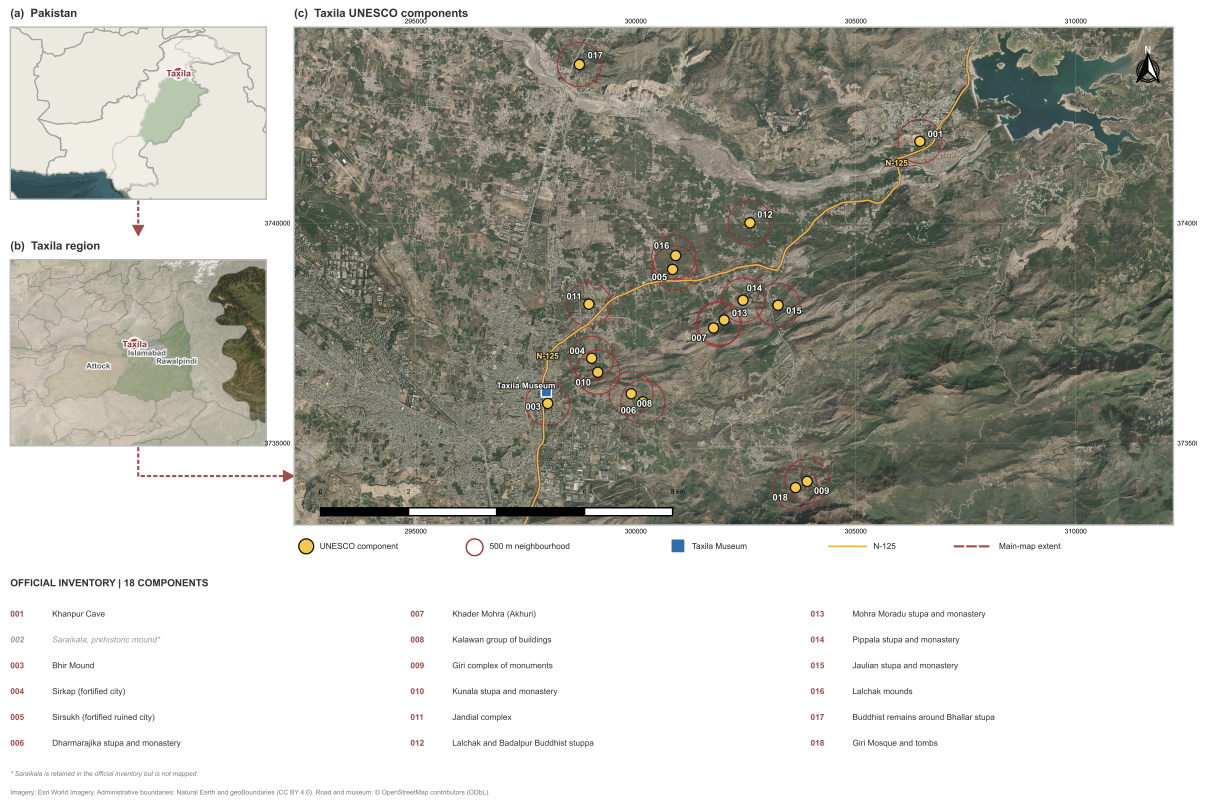


Figure 1: Taxila study-area context and the official 18-component inventory. Seventeen components are mapped; Saraikala is retained in the inventory without an inferred coordinate. Dark-red circles indicate 500 m analytical neighbourhoods. Esri World Imagery provides contextual imagery only; it was not used to calculate spectral, climate or terrain indicators. Boundaries: Natural Earth and geoBoundaries; road and museum: OpenStreetMap contributors (ODbL). Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Table 1: Evidence sources and permitted analytical roles.

Source	Support	Analytical role
Landsat C2 L2	55 scenes; five epochs; 30 m grid	Relative spectral pressure and endpoint convergence
Elevation tile	Approximately 30 m	Terrain and drainage susceptibility
Open-Meteo	Daily, 1991–2025; four cells	Regional trends and acquisition-window context
NASA POWER	Daily, 1991–2025	Weather-product sensitivity
WorldCover consensus	2020/2021 labels matched to E2019 features	Weakly supervised land-cover benchmark
UNESCO component inventory	18 records; 17 mapped points	Component identifiers and neighbourhood centres

preserve acquisition dates, sensor identities and quality information. Cross-sensor and masking limitations were considered explicitly [12, 13].

Daily Open-Meteo ERA5-Seamless data covered 1991–2025 [14]. Component queries resolved to four unique weather cells, each containing 12,784 daily records. NASA POWER supplied a comparison product over the same period [15]. Neither was treated as station truth. Terrain came from the one-arc-second N33E072 elevation tile distributed through AWS Terrain Tiles. ESA WorldCover 2020/2021 consensus [16] supplied six proxy land-cover classes for an E2019 feature table. Table 1 summarises the role of each source. Figure 2 separates the paths through which these inputs inform interpretation.

3.2 Comparable spectral evidence

Cloud, shadow and invalid observations were masked before compositing on a fixed 30 m grid. Endpoint change used the intersection of valid E2004 and E2024 cells, preserving a common denominator. NDVI, NDBI and MNDWI were calculated from scaled reflectance using their standard band ratios [17, 18]. Lower NDVI and MNDWI and higher NDBI were oriented towards greater relative pressure. NDBI can respond to bare soil as well as built surfaces; its direction was a screening assumption, not an attribution of urban expansion.

Within each radius, component-epoch medians were transformed to empirical percentiles using the pooled component-epoch distribution. Denote the oriented percentiles by z_N, z_B, z_W . Correlated vegetation and built/bare terms were grouped before combination:

$$L = \frac{1}{2}(\frac{1}{2}z_N + \frac{1}{2}z_B) + \frac{1}{2}z_W. \quad (1)$$

This structure reduces the influence that two closely related indicators would otherwise receive in an equal three-term sum. It does not remove all statistical dependence. Percentiles are conditional on the analysed inventory and epochs; they are not transferable physical thresholds.

For endpoint changes, the lower 20% tails of NDVI and MNDWI and the upper 20% tail of NDBI defined three adverse criteria. Their sum identified cells meeting at least two or all three criteria. Because these thresholds are data-derived and the indicators correlated, convergence measures descriptive screening prevalence rather than damaged area or statistical hotspot significance. Scene provenance and preprocessing specifications accompany the analysis.

3.3 Terrain, priority and uncertainty

Slope, a contributing-area wetness proxy and distance to a thresholded D8 drainage network described terrain context [19, 20]. Higher slope and wetness and shorter drainage distance were oriented towards greater susceptibility. Their component percentiles were averaged to obtain T .

CHIP Evidence to field-inspection priorities

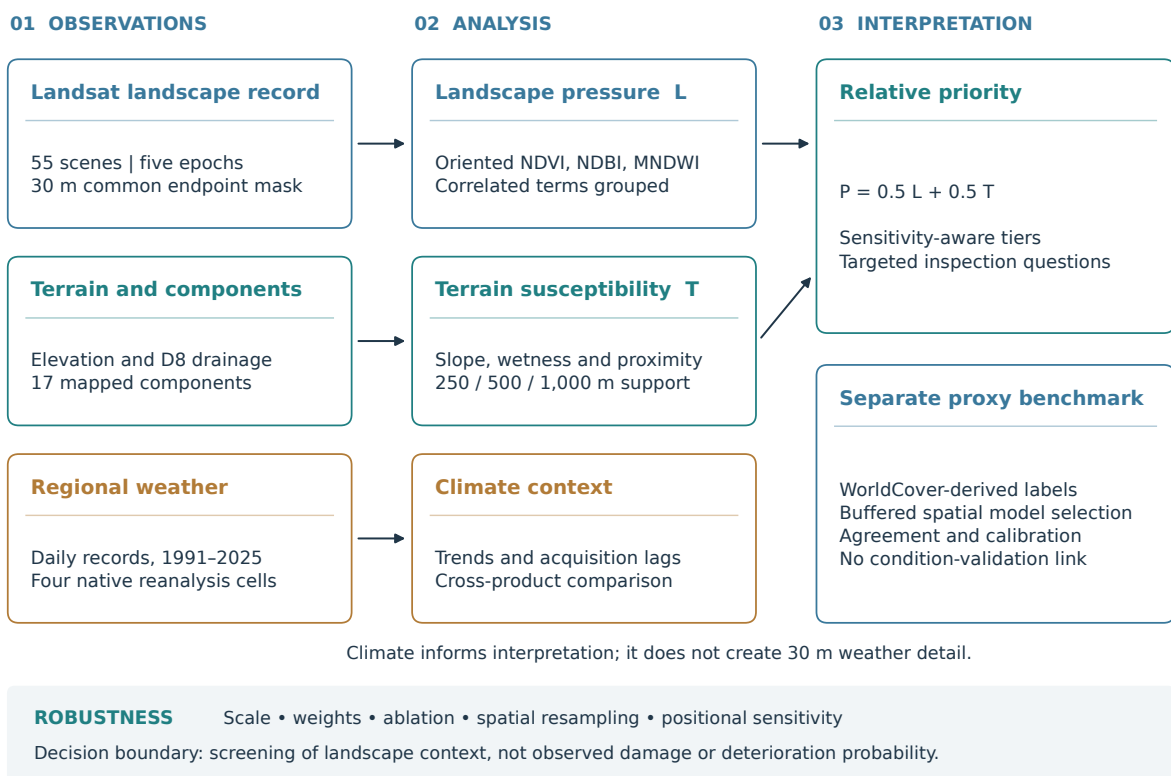


Figure 2: CHIP framework. Landscape and terrain evidence contribute to relative priority; climate retains its contextual role at native support. The land-cover benchmark is a separate diagnostic and does not validate the conservation score. The vector diagram is generated from an editable Python script. Codex assisted schematic preparation; the diagram contains no generated observational imagery.

Flow accumulation represents topographic convergence, not discharge; the DEM cannot resolve all culverts, walls or local drains.

The primary score was $P = 0.5L + 0.5T$. Equal domain weights were a declared structural choice, not an expert-elicited preference or fitted conservation model. Climate was not used to create fine-scale rank differences. A separate integrated context score was retained as a sensitivity analysis. Scores are dimensionless and relative to the same 17 mapped components.

3.4 Experimental assessment of inspection priorities

The experiments examine complementary aspects of the decision rule. First, the hierarchical score was compared with an ungrouped equal-factor mean and landscape-only and terrain-only baselines. Leave-one-factor-out scores were recomputed with retained weights renormalised. Spearman correlation, Kendall concordance, top-five overlap and maximum rank movement quantify departures from the reference; the component ranked first identifies their management consequence.

Second, the 250, 500 and 1,000 m neighbourhoods were crossed with landscape weights of 0.25, 0.50 and 0.75 and three terrain formulations: equal factors, inverse-correlation-weighted factors, and a reduced slope–drainage formulation. The resulting 27 scenarios assess structural persistence. Components occurring in the top three in at least half the scenarios define a core inspection tier; components outside that tier occurring in the top five in at least half define a conditional tier. These operational thresholds organise scenario frequencies and do not represent measured damage severity. A continuous weight experiment sampled $w \sim \text{Beta}(\alpha, \alpha)$ in $P_w = wL + (1 - w)T$, with $\alpha = 0.5, 1, 2$ and 50,000 draws per setting.

Third, 2,000 spatial draws were generated at each of 90, 150 and 300 m block sizes within the 500 m neighbourhoods. Exponential block weights were shared across components occupying the same grid blocks and multiplied by valid-cell counts. Support-weighted medians of block-level spectral medians supplied each draw, with terrain fixed. This design represents spatial dependence and overlap between analytical neighbourhoods.

Fourth, component centres were displaced independently within uniform-area discs of maximum radius 30, 60 or 120 m. Spectral and terrain summaries were re-extracted and percentiles recalculated for 2,000 draws per radius. An undisplaced re-extraction provides the control. It is distinguished from the fixed summary-table reference because feature reconstruction itself can change ranks; differences between these controls must not be attributed to positional error.

Finally, 500 spatial states combined a sampled block size, shared block weights and independently displaced centres within 120 m. Each state was crossed with ten decision settings, sampling a uniform landscape weight and one of the three terrain formulations. The resulting 5,000 score vectors quantify joint sensitivity; they are nested simulations, not 5,000 independent geographic observations. Median ranks, central 95% intervals and top-three frequencies summarise the conditional distributions. Figures 5 and 6 report the central spatial, structural and joint experiments in the main article.

3.5 Climate trends and acquisition windows

The regional series averages each day equally across the four unique weather cells before annual aggregation; it is not an average over 17 independent component climates. Annual metrics were characterised by Theil–Sen slopes [21]. For annual uncertainty analysis, we fitted a robust linear trend, resampled five-year blocks of its residuals and restored the fitted trend to obtain slope intervals. Null slopes were calculated by adding the same resampled residuals to a constant baseline. The two-sided Monte Carlo probability was $(1 + \#\{|\beta_b^0| \geq |\hat{\beta}|\}) / (B + 1)$, with $B = 2,000$. Benjamini–Hochberg adjustment was applied to the 12 annual metrics [22]. The procedure assumes that detrended residual blocks adequately represent short-range dependence; it does not resolve product bias.

Antecedent weather was calculated for every Landsat acquisition over 7, 30, 90, 180 and 365 days and then averaged within each epoch. Product comparisons used matched dates and periods. Weather-spectral correlations involved only five independent composite epochs and were retained as exploratory, without causal interpretation. No weather field was downsampled to the Landsat grid.

3.6 Retrospective spatial proxy benchmark

The feature table contained 20,749 class-stratified samples with six reflectance bands and four indices. The outer split assigned 4,188 samples in block columns 3–4 to testing, excluded 4,467 samples in columns 2 and 5 as a buffer, and retained 12,094 development samples. The 20 test blocks did not overlap the 60 development blocks. The minimum observed sample separation was calculated from grid coordinates. Labels represent WorldCover agreement and carry temporal mismatch with the E2019 predictors.

The experiment compared multinomial logistic regression, random forest, extra trees, histogram gradient boosting, a multilayer perceptron, XGBoost and CatBoost [23–26]. Each family received three predeclared configurations and the same three contiguous inner validation bands. Adjacent block rows were excluded from inner training. Standardisation was fitted inside each training fold; early stopping was disabled to avoid an implicit random validation split. Mean inner macro-F1, computed over all six classes even when a fold omitted a class, selected the primary family and configuration before outer predictions were generated.

The water proxy was geographically concentrated and absent from one inner training fold. We retained this difficult fold, assigned zero predicted mass to an unseen class where necessary, and kept the six-class scoring denominator. This tests transfer under the observed label geography rather than manufacturing class coverage. It also limits the precision of comparisons.

Outer metrics included macro-F1, fixed-six-class mean recall, overall agreement, log loss, Brier score and ten-bin top-label calibration error. Two thousand shared test-block resamples produced paired macro-F1 differences and percentile intervals. A scalar temperature was estimated from development out-of-fold predictions and applied to outer probabilities. Fixed-capacity bands-only, indices-only, location-only, shuffled-label and class-prior controls diagnosed information content; three seeds assessed the selected model’s refit sensitivity. These controls did not alter primary selection.

The outer stripe had been inspected in earlier analyses. The comparison is therefore retrospective, even though its model selection uses development data only. It does not provide a new untouched test, a population-area accuracy estimate, or independent evidence of monument condition [27, 28].

3.7 Computational implementation

The land-cover experiments were executed in Google Colab and independently repeated on Windows with recorded package versions. Model selection was completed before outer predictions were generated. Colab outputs supply the main model comparison. The accompanying code records input identifiers, spatial splits, configurations and per-sample probabilities, supporting reproduction from the distributed derived-data package [29, 30]. Source acquisition and preprocessing are documented separately through scene manifests and provider specifications.

4 Results

4.1 Multitemporal landscape pressure and spectral convergence

The five epochs revealed heterogeneous landscape-pressure patterns across the component neighbourhoods (Fig. 3). Their common comparison support was substantial: 386,280 of 403,480 cells

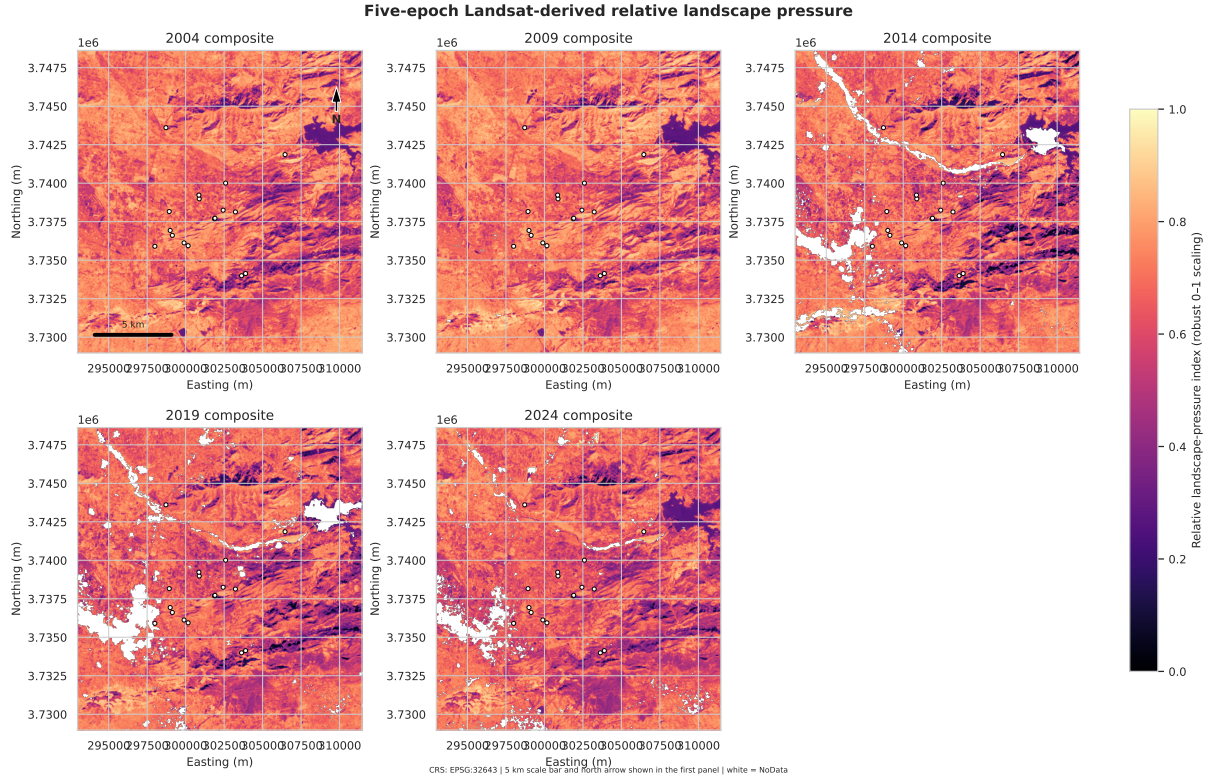


Figure 3: Landscape-pressure patterns across the five Landsat epochs. The common colour scale describes relative pressure; the maps permit comparison of spatial persistence and redistribution across epochs.

(95.74%) were valid at both endpoints, leaving 17,200 cells outside change denominators. On this intersection, 65,391 cells (16.93%) met at least two adverse criteria and 5,493 (1.42%) met all three. The empirical thresholds were $\Delta\text{NDVI} \leq -0.00543$, $\Delta\text{MNDWI} \leq -0.03743$ and $\Delta\text{NDBI} \geq 0.00835$. Figure 4 locates their convergence relative to the wider pressure pattern.

At 500 m, NDVI- and NDBI-pressure ranks were strongly aligned ($\rho = 0.972$). Their grouping therefore affects how surface-cover information contributes to the hierarchy. Giri’s surface-cover term was low (0.0176), while its MNDWI-pressure percentile was 0.9941 and terrain score 0.6765. Its leading composite priority consequently arose from the moisture-related spectral and terrain terms, rather than agreement among every indicator. This decomposition gives the priority an explicit inspection rationale.

4.2 Baselines and ablations identify the drivers of priority

Giri complex led the fixed 500 m score (0.5912), followed by Giri Mosque and tombs (0.5725) and Jaulian (0.5676). The ungrouped equal-factor baseline retained only three of the reference top five, with $\rho = 0.844$ and a maximum movement of seven ranks; it placed Bhallar first. Landscape-only scoring retained one of the top five ($\rho = 0.325$; maximum movement 11) and selected Sirkap, whereas terrain-only scoring retained four ($\rho = 0.830$; maximum movement seven) and selected Bhallar.

Factor ablations were markedly asymmetric (Fig. 6a). Removing NDVI or NDBI retained $\rho = 0.991$ and 0.990 , respectively, and left Giri first. Removing MNDWI instead produced $\rho = -0.280$, no top-five overlap and a maximum movement of 13 positions. Removing slope or drainage proximity produced $\rho = 0.611$ and 0.454 and placed Sirsukh first. The hierarchy thus reduced duplication between the vegetation and built/bare terms, while the moisture-related indicator and terrain pathways remained consequential to the decision. These results identify

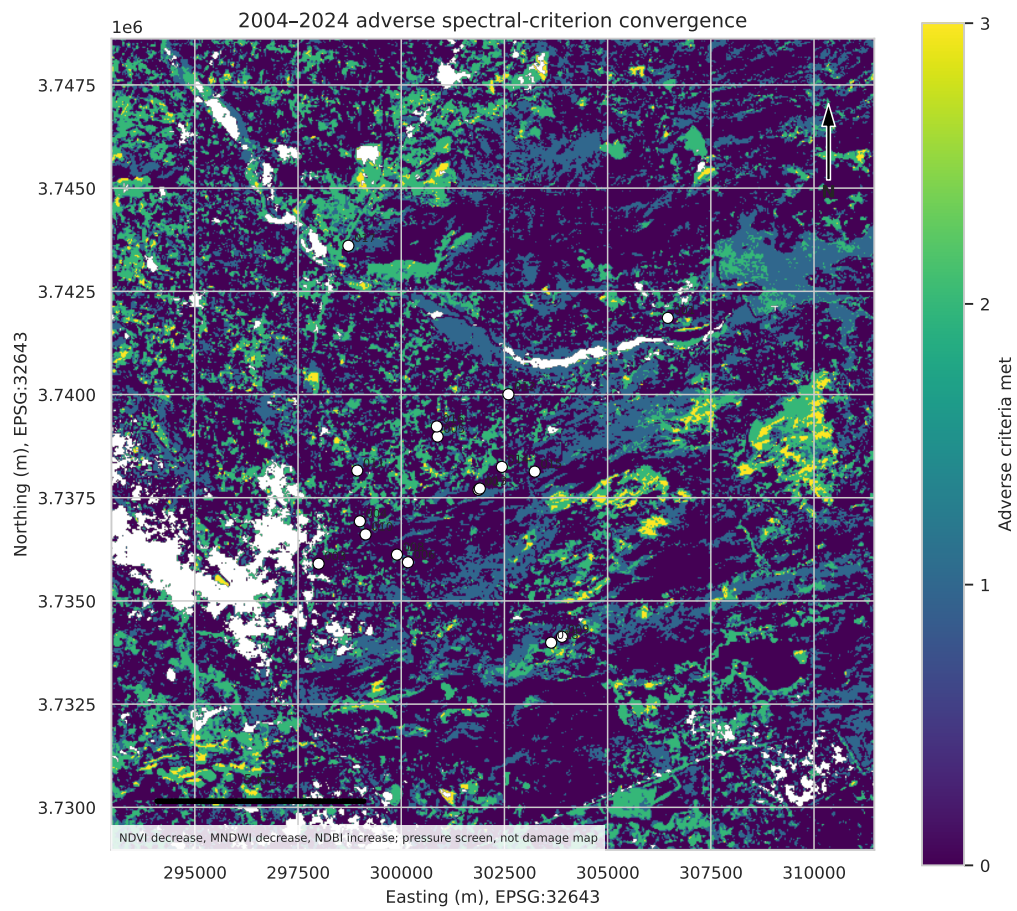


Figure 4: Coincident adverse spectral criteria between E2004 and E2024. Counts use the 386,280-cell common endpoint support. Convergence is a landscape-screening indicator rather than a mapped observation of archaeological damage.

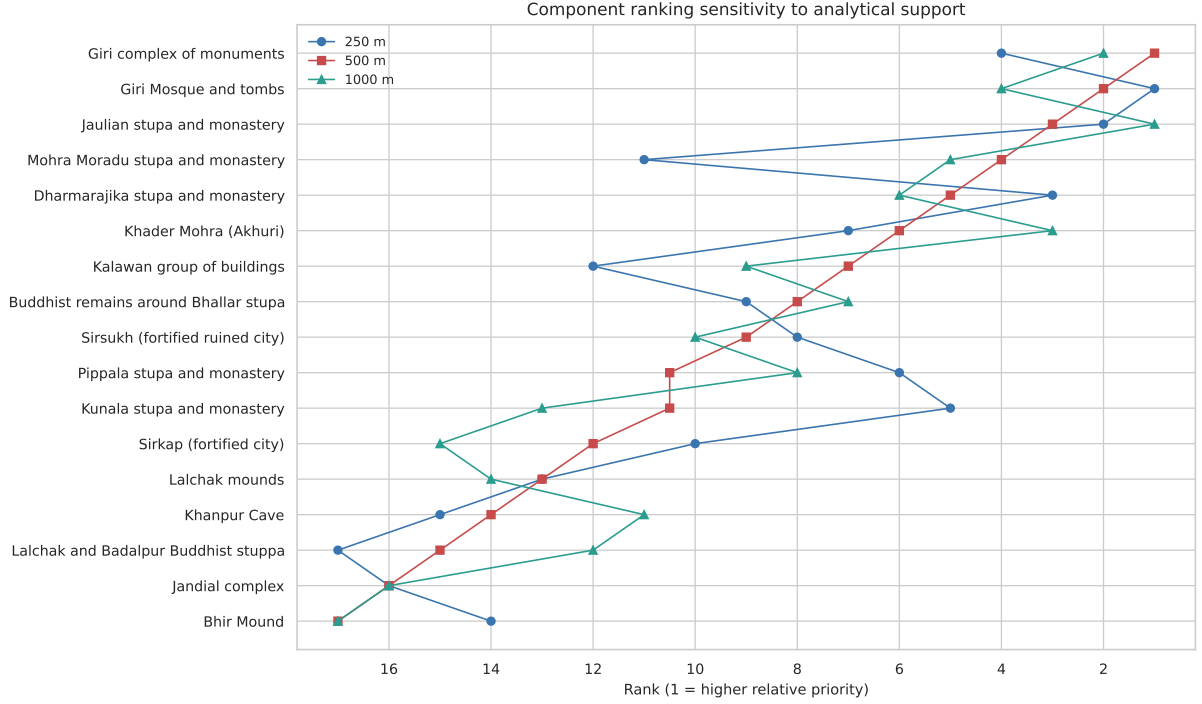


Figure 5: Component ranks at 250, 500 and 1,000 m analytical support. Each line follows one component, exposing the dependence of priority on neighbourhood radius.

where independent observations would most strongly test the proposed inspection order.

4.3 Spatial support, structural choices and inspection tiers

Rank agreement was stronger between 500 and 1,000 m ($\rho = 0.918$) than between 250 and 1,000 m ($\rho = 0.711$; Fig. 5). Jaulian ranked 2, 3 and 1 at the three radii, whereas Giri complex ranked 4, 1 and 2. Giri Mosque and tombs moved from first at 250 m to fourth at 1,000 m. These changes indicate that local and broader landscape context can support different inspection sequences.

Across the 27 structural scenarios, Giri complex, Giri Mosque and tombs, and Jaulian formed the core tier, appearing in the top three in 17, 20 and 18 scenarios, respectively. Khader Mohra and Mohra Moradu formed the conditional tier, appearing in the top five in 15 and 17 scenarios. Scenario agreement therefore supports a small inspection set more clearly than a unique ordering. Continuous domain-weight perturbation reinforced this distinction: Giri's top-three frequency increased from 0.683 to 0.869 as α increased from 0.5 to 2, concentrating weights towards an even balance. Bhallar's frequency decreased from 0.371 to 0.221. The relative emphasis on landscape and terrain is therefore a substantive conservation decision.

4.4 Spatial, positional and joint uncertainty

Shared-block resampling gave Giri top-three frequencies of 0.994, 0.971 and 0.960 for 90, 150 and 300 m blocks (Fig. 6b). Bhallar's frequencies were 0.914, 0.931 and 0.815, while Giri Mosque and tombs declined from 0.773 to 0.390. Larger blocks therefore altered uncertainty unevenly across components. A single block size would obscure this dependence on the spatial organisation of the input evidence.

The undisplaced re-extraction control ranked Bhallar first, Giri second and Jaulian third, demonstrating a difference from the fixed summary-table ordering before positional perturbation. Relative to that control, displacement effects were modest for Bhallar and Giri: at 120 m their

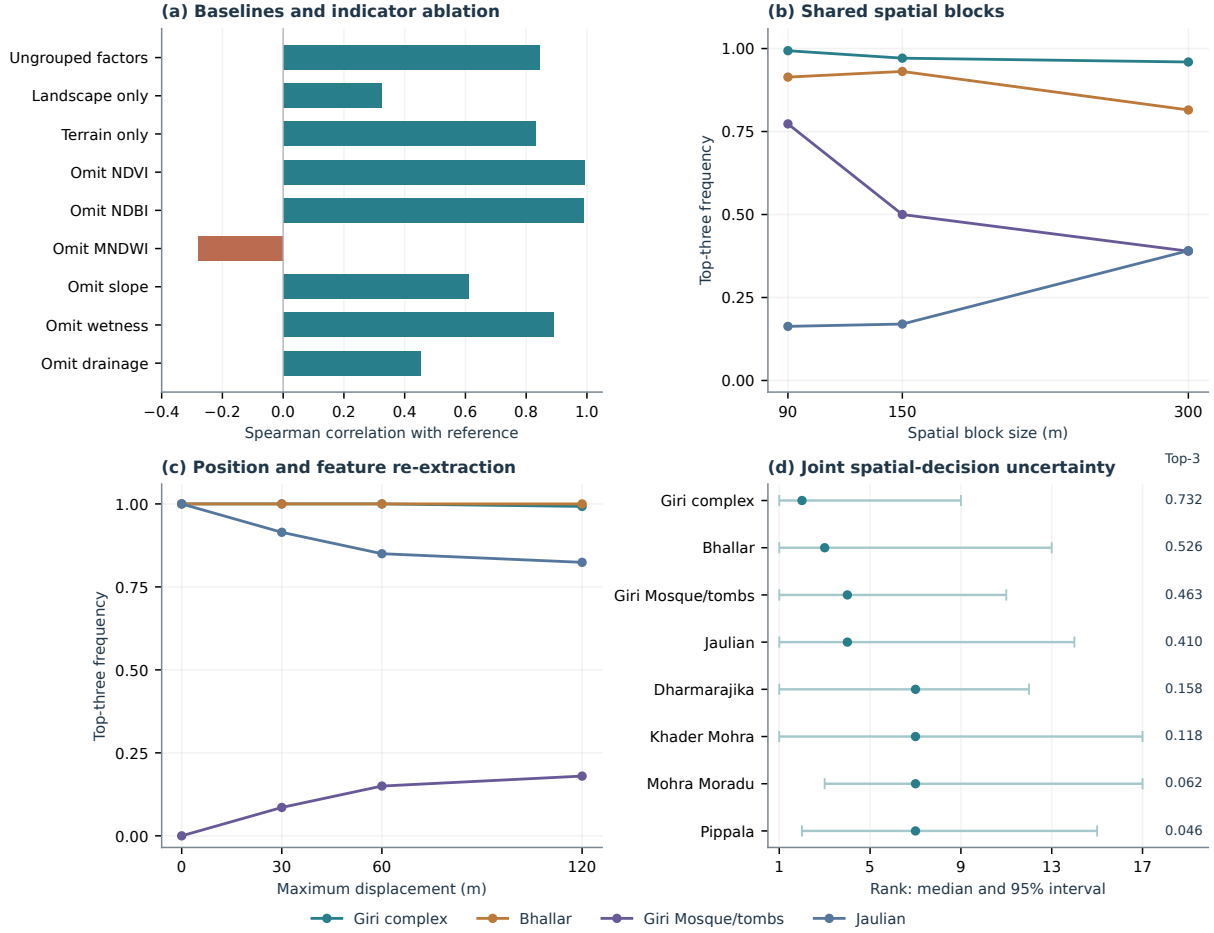


Figure 6: Experimental assessment of the inspection rule: (a) rank concordance after baseline or factor changes; (b) shared-block sensitivity; (c) coordinate-displacement sensitivity with both spectral and terrain re-extraction; and (d) median ranks and central 95% intervals under joint spatial-decision simulation. Labels in (d) give top-three frequencies. Panels (b)–(d) condition on their respective reconstruction and perturbation designs; the zero-displacement control in (c) is distinct from the fixed summary-table reference in (a).

top-three frequencies were 1.000 and 0.993. Jaulian declined from 0.915 at 30 m to 0.824 at 120 m, while Giri Mosque and tombs increased from 0.086 to 0.180 (Fig. 6c). Feature reconstruction and positional uncertainty thus make separable contributions and should not be interpreted as the same experiment.

Joint simulation widened the decision envelope (Fig. 6d). Giri had median rank 2, interval 1–9 and top-three frequency 0.7324; Bhallar had median rank 3, interval 1–13 and frequency 0.5260. Giri Mosque and tombs and Jaulian had frequencies 0.4626 and 0.4096. In particular, Bhallar’s prominence under spatial reconstruction contrasted with its limited top-three persistence across the fixed structural scenarios (2/27). This identifies a component requiring a mechanism-led drainage and terrain inspection, rather than treating one aggregate frequency as a universally stable priority.

4.5 Climate trends and acquisition-matched conditions

Open-Meteo annual precipitation decreased by 10.30 mm year⁻¹ (95% residual-block interval -17.91 to -2.67; annual-family $q = 0.027$), while annual mean temperature increased by 0.0418 °C year⁻¹ (0.0099–0.0731; $q = 0.026$). Heavy-precipitation days declined by 0.143 days year⁻¹ ($q = 0.018$). Soil moisture also declined, although these inferences remain conditional on

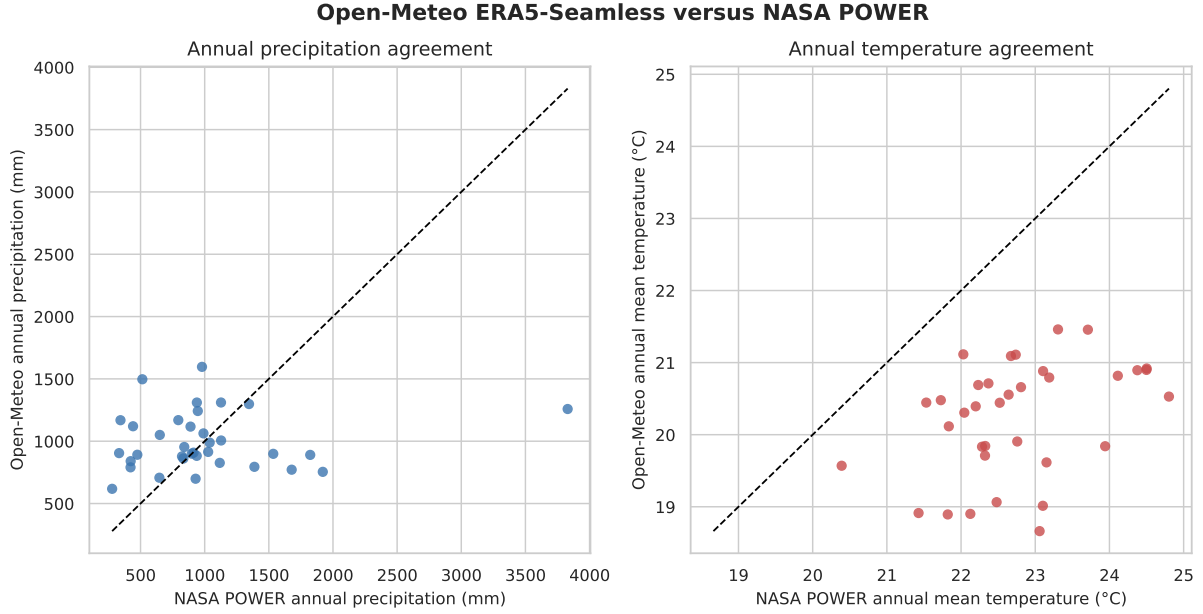


Figure 7: Matched annual precipitation and temperature in the two gridded weather products. Dashed lines indicate equality; product disagreement constrains environmental interpretation in the absence of station validation.

Table 2: Seven-family spatial comparison. The primary model is selected using buffered development folds; outer intervals use 2,000 shared test-block resamples.

Model	Inner F1	Outer F1	95% interval	ECE
Random forest (primary)	0.695	0.821	0.691–0.835	0.014
Logistic regression	0.655	0.807	0.680–0.822	0.050
MLP	0.624	0.849	0.723–0.888	0.020
Extra trees	0.617	0.826	0.694–0.840	0.029
XGBoost	0.616	0.827	0.699–0.845	0.017
CatBoost	0.614	0.792	0.680–0.825	0.017
Histogram boosting	0.611	0.846	0.704–0.857	0.051

the product and temporal dependence model.

NASA POWER gave opposing point slopes: $+29.21 \text{ mm year}^{-1}$ for precipitation and $-0.0372 \text{ }^{\circ}\text{C year}^{-1}$ for temperature. Annual precipitation agreement across 35 years was weak (Spearman $\rho = 0.083$; RMSE $646.74 \text{ mm year}^{-1}$; Fig. 7). These differences materially change the climatic explanation that might otherwise be attached to the landscape maps.

Acquisition-window matching further differentiated the epochs. E2024 antecedent precipitation averaged 42.47, 221.19 and 852.92 mm over 30, 90 and 365 days. The maxima occurred instead in E2004, E2019 and E2014, respectively. Thus the weather preceding the retained satellite scenes cannot be represented reliably by assigning a single wet or dry label to an epoch. Detailed event definitions and annual trends are supplied in Supplementary Table S2.

4.6 Seven-family spatial comparison and diagnostic experiments

Buffered development validation selected random forest with mean macro-F1 0.695. On the 4,188 outer-stripe samples, it achieved macro-F1 0.821 (95% test-block interval 0.691–0.835), agreement 0.813, log loss 0.498 and calibration error 0.014 (Table 2). The MLP achieved the largest outer macro-F1 (0.849), but its lower development score did not support selecting it as the primary model.

Table 3: Feature controls and probability calibration on the same outer stripe. Feature controls use fixed-capacity logistic regression, except the class-prior control. Temperature is fitted exclusively to development out-of-fold probabilities.

Experiment	Macro-F1	Log loss	ECE
Random forest, raw	0.821	0.498	0.014
Random forest, temperature scaled	0.821	0.574	0.136
Bands only	0.824	0.562	0.051
Indices only	0.671	0.816	0.034
Location only	0.140	1.912	0.147
Shuffled development labels	0.070	1.712	0.063
Class prior	0.049	1.706	0.041

Paired spatial resampling showed random forest exceeding logistic regression by 0.0145 macro-F1 (95% interval 0.0028–0.0283). Histogram gradient boosting exceeded random forest by 0.0248 (0.0117–0.0663), whereas the MLP comparison had an interval spanning zero. These unadjusted comparisons expose a difference between development selection and performance on the evaluated stripe; no family dominates every criterion.

The feature controls provide a complementary interpretation (Table 3). Fixed-capacity logistic regression using bands alone reached macro-F1 0.824, whereas indices alone reached 0.671. Location-only, shuffled-label and class-prior controls achieved 0.140, 0.070 and 0.049. Spectral information therefore explained substantially more agreement than coordinates or these null controls, but the selected ten-feature random forest did not outperform the simpler bands-only control by its point estimate. This limits the incremental predictive claim for the more elaborate representation.

Development-only temperature scaling ($T = 1.658$) increased outer log loss from 0.498 to 0.574, Brier score from 0.269 to 0.300 and calibration error from 0.014 to 0.136. The calibration learned from geographically separated folds transferred poorly to the outer stripe. Random-forest macro-F1 ranged only from 0.8205 to 0.8214 across three refit seeds and matched between Colab and Windows, indicating that the larger differences arose from modelling and spatial-evaluation choices rather than the selected model’s refit seed.

5 Discussion

5.1 From environmental indicators to inspection decisions

The principal contribution is the experimental evaluation of an inspection rule across several distinct sources of uncertainty. A component’s score is accompanied by evidence about its drivers, its persistence across support and structural choices, and its response to spatial reconstruction. This extends the practical value of a priority map: an inspection team can identify both where to visit and which environmental explanation requires verification.

The ablations show why the hierarchy matters. NDVI and NDBI supplied closely aligned information, while MNDWI and terrain pathways strongly affected the ordering. Giri’s leading score was consequently associated with moisture-related spectral and terrain terms despite low surface-cover pressure. An appropriate field visit would investigate drainage routes, surface moisture and their relationship to the monument fabric, while a landscape-led priority such as Sirkap would motivate examination of surrounding surface-cover change. These are testable inspection questions rather than interchangeable meanings of a high score.

The results complement Taxila research combining ground observations and satellite evidence [4], and wider heritage multi-hazard assessments [3, 31]. CHIP adds a structured way to test how priorities respond to analytical choices before those priorities are operationalised. The

transferable contribution is the experimental procedure; the Taxila percentiles, weights and tier thresholds require local reassessment at another property.

5.2 Interpreting persistence and disagreement

Giri, Giri Mosque and tombs, and Jaulian persisted across the structural scenarios, supporting a core inspection set. However, Bhallar illustrates why scenario frequencies must retain their definitions. It was seldom in the fixed structural top three but prominent after feature re-extraction and under spatial simulation. Combining those results into one unconditional confidence score would conceal the information most relevant to planning: Bhallar’s priority depends strongly on how its local terrain and spectral support are represented.

Joint intervals are wider than the certainty implied by a single ranking. They justify combining a persistent core set with additional visits targeted to particular mechanisms and representation-sensitive components. Access, cultural significance, repair history and specialist assessment would then guide the operational sequence. Those management factors should enter as explicit decisions rather than being inferred from remotely sensed variables.

5.3 What the predictive experiments establish

The seven-family comparison demonstrates useful geographic land-cover agreement, while the controls reveal where extra complexity adds little. Bands-only logistic regression remained competitive with the selected random forest, and temperature scaling worsened probability quality. Development selection and outer performance also diverged. Reporting these experiments together is more informative for model choice than selecting the largest test score: the appropriate model depends on spatial transfer, uncertainty and the intended output, not only point accuracy.

These findings are consistent with the importance of spatial evaluation and applicability domains [8–10]. Recent tabular foundation models offer additional comparative opportunities [32], but the present target remains consensus land cover. Class-stratified sampling, temporal mismatch and geographically concentrated water labels constrain interpretation. The previously examined outer stripe makes this a retrospective comparison, and neither it nor a more capable classifier provides independent monument-condition validation.

5.4 Climate interpretation and study limitations

Opposing product trends prevent a simple attribution of the landscape pattern to regional drying or warming. Acquisition-matched weather describes conditions relevant to the observations more directly, but five composite epochs provide limited temporal replication. Station measurements, soil observations and higher-frequency imagery would help distinguish climatic responses from management, seasonality and land-use processes [5, 33].

Point-centred neighbourhoods approximate landscape context rather than monument footprints; Saraikala lacks usable geometry. DEM routing cannot resolve every local obstruction, and percentile scores depend on the analysed inventory. Spatial simulations quantify stated perturbations without covering all label, sensor or conservation-model error. The next validation should therefore use independently assessed component-condition observations, record assessor agreement and evaluate components or periods excluded from design choices. Such evidence would test whether the proposed inspection strategy identifies actionable conservation problems.

6 Conclusions

CHIP links heterogeneous geospatial observations to inspection priorities through an explicit hierarchy and a coordinated experimental assessment. At Taxila, baseline comparisons and

ablations showed that moisture-related spectral information and terrain pathways controlled important rank changes. Multiscale and structural scenarios identified a persistent core inspection set, while spatial reconstruction and joint simulations exposed additional, mechanism-dependent priorities. Acquisition-matched weather and product comparison constrained climatic interpretation. The separate spatial land-cover experiments demonstrated useful proxy agreement but also competitive simple controls and poor calibration transfer. Together, these results support a component-specific inspection programme with documented uncertainty, providing a practical basis for subsequent field validation and adaptation to other archaeological landscapes.

Data and code availability

The review package includes derived evidence, source manifests, executable notebooks, split records, configurations and result checksums. Original third-party products remain subject to provider terms. Identifying repository details are withheld from the anonymised manuscript and supplied separately to the editor. The supported computational workflow begins from the documented derived-data package; upstream acquisition and preprocessing requirements are described in the source manifests.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI ChatGPT/Codex to assist with code development and review, experiment execution, reproducibility checks, schematic preparation and manuscript drafting and editing. The authors are responsible for reviewing the code, numerical results, references and interpretations and for approving the final submission. The tool was not treated as an author or as scientific evidence.

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